



Bartholomew Richard

Research Assistant
Department of Chemistry & Biological Sciences and Engineering
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EDUCATION

Degree/Certificate	Institute/Board	CGPA Percentage	Year
M. Sc. in Chemistry	Fatima Mata National College (Autonomous), (Affiliated with the University of Kerala)	7.9	2020-2022
B. Sc. in Chemistry	Fatima Mata National College (Autonomous), (Affiliated with the University of Kerala)	8.7	2017-2020

TECHNICAL SKILLS

- **Software:** Chemdraw, Biorender, Blender, Microsoft Office Suite: Excel, Word, PowerPoint.
- **Laboratory:** Techniques like soft template (Liquid-liquid interface), hydro/solvothermal, etc. Synthesis of MXenes, transition metal oxides (TMOs), transition metal dichalcogenides (TMDs), metal-organic framework, etc., conducting polymers, hydrogels, metal organic gels, metallogels.
- **Characterization:** Electrochemical (Cyclic Voltammetry, Chronoamperometry, Differential Pulse Voltammetry, Mott-Schottky Analysis), Ultraviolet Photoelectron Spectroscopy (UPS), X-ray Photoelectron Spectroscopy (XPS), Raman spectroscopy, Scanning Electron Microscopy, XRD, UV-Visible Spectroscopy.

PUBLICATIONS

- **Richard Bartholomew**, Manjusha Mathew, T. Janani, Jithin Thomas Chacko, Kannankuzhiyan Niyas, Priyadharishini Veeraraghavan, P Abdul Rasheed, "Silane-Coupling Impelled Metal Oxide-MXene Proximity: An Electrochemical Probe for Tracking *enteral histaminosis* In Vitro." **2025 (Manuscript Submitted)**
- **Richard Bartholomew**, K Niyas, Manjusha Mathew, P Abdul Rasheed, "Lab on a syringe: Strategic design of 1T SnS₂/MXene embedded single-stranded carbon yarn for the real-time stress biomarker detection." [*ACS Applied Electronic Materials*, 2024, 6, 9, 6905–6915.](#)
- **Richard Bartholomew**, K Niyas, Menon Ankitha, P Abdul Rasheed, "Biological Metal-Organic Framework-Embedded MXene Nanocomposite as a Wearable Transducer Patch for Real-Time Monitoring of the Sleep Hormone." [*ACS Applied Nano Materials*, 2024, 7, 8, 9585–9597.](#)
- **Richard Bartholomew**, C Shahana, Raju Vivek, Amarendar Reddy, P Abdul Rasheed, "Acoustic platforms meet MXenes – a new paradigm shift in the palette of biomedical applications." [*Nanoscale*, 2023, 15, 18156–18172.](#)
- **Richard Bartholomew**, Susmi Anna Thomas Amarendar Reddy, Mohan Reddy Pallavolu, Jayesh Cherusseri, "Minireview on Fluid Manipulation Techniques for the Synthesis and Energy Applications of Two-Dimensional MXenes: Advances, Challenges, and Perspectives." [*ACS Energy and Fuels*, 2023, 37, 10, 6999–7013.](#)
- K Niyas, **Richard Bartholomew**, Menon Ankitha, Pathath Abdul Rasheed, "Sea urchin nanostructured nickel cobaltite modified carbon cloth integrated wearable patches for the on-site detection of the immunosuppressant drug mycophenolate mofetil." [*Analyst*, 2024, 149, 3615–3624.](#)

RESEARCH EXPERIENCE

Indian Institute of Technology Palakkad

Palakkad, India

Research Assistant

March 2023 – July 2025

- Investigating the electrochemical properties of MXene heterostructures (with transition metal oxides, transition metal dichalcogenides, MOF, etc.) for biosensing applications.

National Institute of Technology Calicut (Collaborative Project)

Palakkad, India

Research Assistant

Sept 2024 – Mar 2025

- Investigating the catalytic role of MXene in interfacial polymerization reaction and its nanostructure in developing electrodes for combating antimicrobial resistance. (*Indian Patent Filed-Application No. 202541065580*)

National Institute of Technology Andhra Pradesh

Tadepalligudem, India

Short-Term Intern

Nov 2022 – Feb 2023

- Investigating modulated hydro-thermal synthesis of Fluorescent Metal-Organic Frameworks for bioimaging applications in cancer studies.

National Institute of Technology Calicut

Calicut, India

Master's Thesis

March 2022 – May 2022

- Developed a new UV-mediated exfoliation technique for the synthesis of MXenes and conducted a comparative study with the existing techniques. Also tried a novel strategy for synthesizing MXene-conducting polymer films.

Indian Institute of Technology Indore

Indore, India

Summer Intern

July 2021

- Reviewed the role of two-dimensional materials in the field of energy materials and explored different characterization techniques.

ACHIEVEMENTS

- Best Poster Award**

Poster Presentation at the International Conference on Sustainable and Applied Nanotechnology for Agriculture and Health (SANTAH) organized by the **Department of Chemical Engineering, IIT Madras, Chennai, India.**

- Best Paper Award**

Oral Presentation Category in the 2nd Indo-Korea Virtual Conference on Development of Advanced Materials for Future Technologies (DAMFT - 2021) jointly organized by the **Korea Advanced Institute of Science and Technology (KAIST), Daejeon, South Korea & Vellore Institute of Technology (VIT), Chennai, India.**

- Best Lecturer Award (Student Category)**

Interdepartmental competition organized by the Department of Chemistry, Fatima Mata National College (Autonomous), Kollam, Kerala on behalf of “**International Year of Periodic Table – 2019**”

REFERENCES

- Dr. Abdul Rasheed P
DBT Ramalingaswami Fellow,
Department of Biological Sciences & Engineering,
Indian Institute of Technology Palakkad.
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- Dr. Amarendar Reddy M,
Assistant professor, Department of Chemistry,
School of Sciences,
National Institute of Technology Andhra Pradesh
Email : amarendar@nitandhra.ac.in